American Computer Science League

2022-2023 ● Contest 2: Short Problems ● Elementary Division Solutions

1. Prefix/Infix/Postfix

7 3 − 9 4 + * 5 2 * − 6 / = (7 − 3) (9 + 4) * (5 * 2) − 6 /
= 4 13 * 10 − 6 / = (4 * 13) 10 − 6 /
= 52 10 − 6 / = (52 − 10) 6 /
= 42 6 / = (42 / 6) = 7

A. 7

2. Prefix/Infix/Postfix

/ − + * 3 7 / 8 2 ^ 5 2 9 = / − + (3 * 7) (8 / 2) (5 ^ 2) 9
= / − + 21 4 25 9 = / − (21 + 4) 25 9
= / − 25 25 9 = / (25 − 25) 9
= / 0 9 = (0 / 9) = 0

E. 0

3. Prefix/Infix/Postfix

$$\frac{6 + 9 * (7 - 5)}{(4 - 2)^3} = (6 + 9 * (7 - 5)) / ((4 - 2) ^ 3)$$

= (6 + 9 * (− 7 5)) / ((− 4 2) ^ 3)
= (6 + (* 9 − 7 5)) / (^ − 4 2 3)
= (+ 6 * 9 − 7 5) / (^ − 4 2 3)
= / + 6 * 9 − 7 5 ^ − 4 2 3

C. / + 6 * 9 − 7 5 ^ − 4 2 3

4. Prefix/Infix/Postfix

(7 * 8 − (9 + 3) / 4) ^ (2 − 1) = ((7 8 *) − (9 3 +) / 4) ^ (2 1 −)
= ((7 8 *) − (9 3 + 4 /)) ^ (2 1 −)
= (7 8 * 9 3 + 4 / −) ^ (2 1 −)
= 7 8 * 9 3 + 4 / − 2 1 − ^

B. 7 8 * 9 3 + 4 / − 2 1 − ^

5. Prefix/Infix/Postfix

4 7 5 3 + + * 6 2 ^ − 8 / = 4 7 (+ 5 3) + * (^ 6 2) − 8 /
= 4 (+ 7 + 5 3) * (^ 6 2) - 8 /
= (* 4 + 7 + 5 3) (^ 6 2) − 8 /
= (− * 4 + 7 + 5 3 ^ 6 2) 8 /
= / − * 4 + 7 + 5 3 ^ 6 2 8

D. / − * 4 + 7 + 5 3 ^ 6 2 8